GHraisoni INTERNATIONAL SKILL TECH UNIVERSITY Pune

G H Raisoni International Skill Tech University, Pune.

Question Paper

Continuous Assessment Examination (CAE)

Programme: B Tech **Year: First** **Semester: I**
Course Code: MT102 **Max Marks: 20**
Course Name: Engineering Physics-I **Duration: 60 mins.**

Instructions:

1. All questions are **compulsory**.
2. Only a Scientific Calculator is allowed, include clear diagrams whenever necessary.
3. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q.No. 2C, etc.

Note: This question paper addresses below Course Outcome (CO):

CO1	Demonstrate a comprehensive understanding of interference, diffraction, and polarization phenomena and gain a mathematical aptitude for solving numericals related to these optical features.

Q. No.	Sub Q. No.	Question	Marks
1	Solve any **TWO**		
	A	Derive the expression for path difference in Young's double slit experiment. Extend it to the conditions of maxima and minima and fringe width.	05
	B	Explain how thin film interference occurs in soap bubbles due to reflected light and derive the condition for constructive interference and destructive interference in this case.	05
	C	**a)** A soap bubble (refractive index n = 1.33) appears bright when viewed in reflected light of wavelength 650 nm. If the angle of reflection is 30°, calculate the thickness of the bubble that leads to constructive interference for the reflected light? **b)** In a double-slit experiment, green light with a wavelength of 5200 Å is used. If the screen is 1.5 m away from the slits, and the distance between two consecutive bright fringes is 0.3 cm, calculate the separation between the slits?	05
	D	A glass wedge of angle 0.01 radian is illuminated by monochromatic light of wavelength 6000 Å falling normally on it. At what distance from the edge of the wedge, will the 10th fringe be observed by reflected light.	05
2	Solve any **TWO**		
	A	Derive expression for maximum and minimum for single slit diffraction. Plot the intensity distribution curve of principal maximum and secondary maxima and minima.	05
	B	Compare the salient features of diffraction and interference. Select any optical feature and explain the application of the feature from an engineering point of view.	05
	C	**a)** A diffraction grating has 3000 lines per cm. Calculate the angle of the first-order maximum for light of wavelength 500 nm. **b)** How many orders of diffraction can be observed for a grating with 4000 lines per cm when illuminated with light of wavelength 700 nm?	05
	D	Describe dichroism and double refraction with the help of a neat labelled diagram.	05